

MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

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Abstract—Copy-paste augmentation is widely applied to improve object detection but is rarely evaluated against the specific failure mode a dataset exhibits. We propose MinImageScorer, an error-driven framework that scores per-object detection difficulty from baseline model failures using a cost function combining prediction confidence and spatial accuracy, with weights auto-calibrated via Pearson correlation optimization. We validate the score on two agricultural benchmarks—MinneApple and Global Wheat Head (GWHD)—and use it to guide copy-paste augmentation. Experiments across three random seeds show that score-guided selection outperforms random selection on MinneApple ($\Delta AP = +0.013$) but neither condition recovers the no-augmentation baseline. Feature-space analysis confirms that augmented images shift the training distribution away from the test distribution, explaining the consistent degradation. On GWHD, domain shift dominates and copy-paste provides no benefit under either selection strategy. Our findings establish that the utility of copy-paste augmentation depends on the dataset’s underlying failure mode, and that MinImageScorer correctly identifies the objects responsible for that failure.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

I. INTRODUCTION

Agricultural object detection benchmarks such as MinneApple [?] and Global Wheat Head (GWHD) [?] present distinct structural failure modes. MinneApple is dominated by high-density small objects near the resolution floor of standard backbones, while GWHD contains images from multiple geographic sources with significant cross-domain visual variation. These failure modes differ fundamentally from the object *underrepresentation* problem that Kisantal *et al.* [?] addressed on MS COCO, where copy-paste augmentation produced measurable gains by increasing small-object anchor matches during training. Ghiasi *et al.* [?] further established large-scale copy-paste as a general augmentation strategy for instance segmentation, but likewise evaluated it on datasets without structural resolution or domain-shift limitations.

A critical gap remains: practitioners routinely apply copy-paste augmentation without first diagnosing whether the dataset’s failure mode is addressable by adding more object instances. When the failure is structural—resolution-limited detectability or cross-domain shift—copy-paste may corrupt the training distribution rather than improve it. No prior work

provides an empirical framework to diagnose this mismatch before augmentation is applied.

We make the following contributions:

- 1) **MinImageScorer**: a per-object difficulty score derived from model failure signals, with weights auto-calibrated via Pearson correlation optimization, requiring no domain-specific assumptions.
- 2) **Empirical validation** that the score captures dataset-specific failure modes on both MinneApple and GWHD.
- 3) **Controlled augmentation analysis** comparing score-guided and random copy-paste across three seeds, with feature-space analysis explaining the observed performance patterns.

II. METHODOLOGY

A. MinImageScorer: Formulation

Let $G(I)$ denote the ground-truth objects in image I and $\mathcal{P}$ the set of model predictions. For object g , let $\mathcal{P}(g; \tau)$ be the predictions satisfying $\text{IoU}(p, g) \geq \tau$. The *object difficulty score* is defined as:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g; \tau)} [\alpha (1 - \text{conf}(p)) + \beta (1 - \text{IoU}(p, g))], & \text{if } \mathcal{P}(g; \tau) \\ \alpha + \beta, & \text{otherwise.} \end{cases} \quad (1)$$

Missed detections receive the maximum penalty $\alpha + \beta$. The *image-level score* aggregates object difficulties and penalizes false positives:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I; \gamma, \tau). \quad (2)$$

Weights α, β, w_1, w_2 are calibrated by maximizing the Pearson correlation between score distributions and per-image miss rate and false-positive rate on the training set. This data-driven calibration requires no domain-specific tuning.

B. Score-Guided Copy-Paste Pipeline

The pipeline proceeds in five steps: (1) train a baseline YOLOv8s [?] on the original training set with all built-in augmentations disabled; (2) run inference on the training set to collect per-prediction confidence and IoU signals; (3) compute

TABLE I
THREE-SEED MEAN COCO AP ON ORIGINAL TEST SETS. **BOLD** = BEST
COPY-PASTE CONDITION PER DATASET.

Dataset	Condition	AP	AP ₅₀	AP ₇₅	AP _S	AP _M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	0.428	0.759	0.440	0.286	0.581
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	0.484	0.905	0.455	0.098	0.477

S_{obj} and S_{img} for all objects and images; (4) sample objects proportionally to their scores using exponential weighting ($\alpha=2$) for score-guided, or uniformly for random selection; and (5) paste 2–8 objects per augmented image within the source image (same-image pasting), then retrain from COCO pretrained weights on the augmented set. Both conditions are identical in all parameters except the selection policy, ensuring a controlled comparison.

III. EXPERIMENTS

A. Setup

Datasets. MinneApple [?] contains [FILL: $N_{\text{train}}/N_{\text{val}}/N_{\text{test}}$] images of apple trees with dense small-object annotations. GWHD [?] contains [FILL: $N_{\text{train}}/N_{\text{val}}/N_{\text{test}}$] images of wheat fields collected across multiple geographic domains (Australia, Japan, India, and others), making it a multi-domain benchmark.

Model and training. All models use YOLOv8s [?] initialized from COCO pretrained weights. Training settings: MinneApple 200 epochs / patience 10; GWHD 100 epochs / patience 8; batch size 16; input size 1024×1024 ; learning rate 10^{-2} . All Ultralytics built-in augmentations (mosaic, mixup, HSV jitter, flips) are disabled to isolate the effect of copy-paste.

Augmentation parameters. Dataset augmentation ratio 0.3; scale jitter [0.9, 1.1] (MinneApple) and [0.8, 1.2] (GWHD); no rotation; overlap avoidance with 5 px margin; no alpha blending. Results are reported as means over seeds {123, 456, 789}.

B. Results

Table ?? reports 3-seed mean COCO AP on original test sets for both datasets.

On MinneApple, score-guided copy-paste outperforms random copy-paste by $\Delta\text{AP}=+0.013$, with consistent gains across AP, AP₅₀, AP₇₅, AP_S, and AP_M. This indicates the score successfully directs augmentation toward harder, smaller objects that provide greater gradient signal. However, both copy-paste conditions remain below baseline (-0.024 for Score CP), showing that the distributional cost of pasting outweighs the selective benefit on this dataset.

On GWHD, score-guided and random selection are statistically indistinguishable (gap: -0.002 AP), and both degrade below baseline (-0.023). Notably, on AP_S the score-guided condition is worse than random (0.098 vs. 0.118), because score-guided sampling concentrates pasted objects from

FIGURE PLACEHOLDER:
Image-level UMAP projections for MinneApple (left) and GWHD (right). Groups: original train, augmented (random CP), augmented (score CP), and test images.

Fig. 1. Image-level backbone feature projections (YOLOv8s P4 layer, global average pool, UMAP). Augmented images occupy regions displaced from the test distribution relative to original training images on both datasets. Score-guided augmented images show greater displacement on GWHD due to domain-minority object concentration.

domain-minority sources (predominantly *arvalis*), amplifying the existing domain imbalance rather than correcting it.

C. Feature-Space Analysis

Fig. ?? shows image-level backbone embeddings (YOLOv8s P4, global average pool, UMAP projection) for four groups: original training images, random-CP augmented images, score-CP augmented images, and test images.

On both datasets, augmented images occupy a region of feature space displaced from the test distribution relative to the original training images. Quantitatively, the centroid distance from augmented images to the test set exceeds that of the original training images: MinneApple: [value] (Score CP) vs. [value] (original train); GWHD: [value] (Score CP) vs. [value] (original train). On GWHD, score-guided augmented images show greater displacement than random augmented images, consistent with domain-minority concentration. This distributional shift directly explains the consistent performance degradation observed in Table ??.

IV. DISCUSSION

Why copy-paste fails on these datasets. Kisantal *et al.* [?] demonstrated that copy-paste improves AP_S on MS COCO by increasing anchor coverage for underrepresented small objects. This benefit presupposes that (1) hard objects are above the backbone feature resolution floor, and (2) pasted objects share the same visual domain as the target scene. Neither condition holds reliably here. On MinneApple, the hardest objects identified by MinImageScorer are predominantly at or below the resolution limit of the backbone, making them weakly represented in feature maps regardless of paste frequency. On GWHD, even within-image pasting fails because the feature-space mismatch originates at the image level. These structural limitations were not present in the COCO setting studied by either Kisantal *et al.* [?] or Ghiasi *et al.* [?].

Diagnostic value of the score. Despite producing no absolute improvement, MinImageScorer demonstrates clear diagnostic utility: the score correlates with per-image miss

rate (r =[value] on MinneApple, r =[value] on GWHD), and on MinneApple it selects objects that yield a +0.013 AP relative gain over uniform random selection. The score identifies the right objects; the limitation is that copy-paste cannot make those objects more learnable when the failure mode is structural.

Boundary condition for practitioners. Score-guided copy-paste provides a positive relative benefit when selected objects are above the resolution floor and within-distribution with the target scene. It provides no benefit when the failure mode is domain shift, and it amplifies AP_S degradation when selected objects are sub-resolution. A practical diagnostic criterion follows: run MinImageScorer, inspect the size and domain distribution of high-score objects, and apply copy-paste augmentation only when those objects satisfy both conditions above.

V. CONCLUSION

We proposed MinImageScorer, a model-failure-driven difficulty scoring framework validated on two agricultural detection benchmarks. Controlled experiments show that score-guided copy-paste outperforms random copy-paste on MinneApple (+0.013 AP) but both strategies degrade below baseline, with the degradation confirmed by feature-space analysis to originate from distributional shift between augmented training images and the test set. On GWHD, domain shift makes copy-paste ineffective under any selection strategy.

The practical implication is that copy-paste augmentation should be preceded by failure-mode diagnosis: it is effective for underrepresentation (COCO-like datasets) but not for resolution-limited or domain-shifted datasets. Future work includes online augmentation to reduce distributional rigidity, and domain-constrained pasting for multi-source benchmarks.

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